PARIKSHA365 — STUDY NOTES

Biology — Common Book

SSC, RRB, State PSC, Banks (GA) — General Science + Health

SSC | RRB | BANKS | STATE PSC

v 2026-04

The promise. Read this book end-to-end (with the usual two re-reads + active recall on the embedded prompts) and you will be able to attempt every quiz question in the Pariksha365 pool for this subject with a strong fundamental grasp.

Before You Begin

Forget memorising isolated facts. Every biology question in SSC / Banks / RRB / PSC comes from one of four buckets:

- **Names & discoverers** — who discovered the cell, the laws of heredity, blood groups.
- **Cause & effect** — disease ↔ vector, hormone ↔ disorder, vitamin ↔ deficiency.
- **Numbers & extremes** — 206 bones, 120-day RBC life, 4-chamber heart, 23 pairs of chromosomes.
- **Comparisons** — prokaryote vs eukaryote, mitosis vs meiosis, monocot vs dicot.

This book hits all four buckets hard, in the order examiners love.

How to read this book:

- **First pass** — read end-to-end for context.
- **Second pass** — cover the right column of every table and self-test.
- **Final pass** — the 25-question mini-mock at the end tells you what you've missed.

Examiner Blueprint — What Actually Appears on Papers

EXAM ALERT

High-yield order (SSC CGL/CHSL, RRB NTPC, IBPS PO GA section):

1. **Vitamins & deficiency diseases** — 2-4 questions per paper, reliably. Know every row of the vitamin table cold.
2. **Disease-vector mapping** — Malaria/Dengue/Filaria vectors are tested in every single exam.
3. **Human body extremes** — Smallest/largest bone, organ, gland, cell. Appears in every paper.
4. **Hormones & glands** — "Master gland", "fight-or-flight", goitre cause — all predictable.
5. **Cell organelles** — "Powerhouse", "suicide bags", "protein factory".
6. **Blood groups** — Universal donor, universal recipient, discoverer.
7. **Plant biology** — Photosynthesis equation, C3 vs C4, fruit-ripening hormone.
8. **Ecology** — 10% rule, biodiversity hotspots, pyramid of energy (always upright).

Topics examiners rarely test: detailed embryology, molecular biology mechanisms, specific enzyme kinetics. Don't over-invest here.

How to read this book — your real study load

Total pages: 37. Don't let the page count scare you.

Mandatory reading — pages 1 to 37 (100% of the book)

That's the full syllabus. Master those 37 pages and you've covered every concept an examiner can fairly ask.

Index — Table of Contents (clickable)

PART A — FOUNDATIONS: CLASSIFICATION & THE CELL

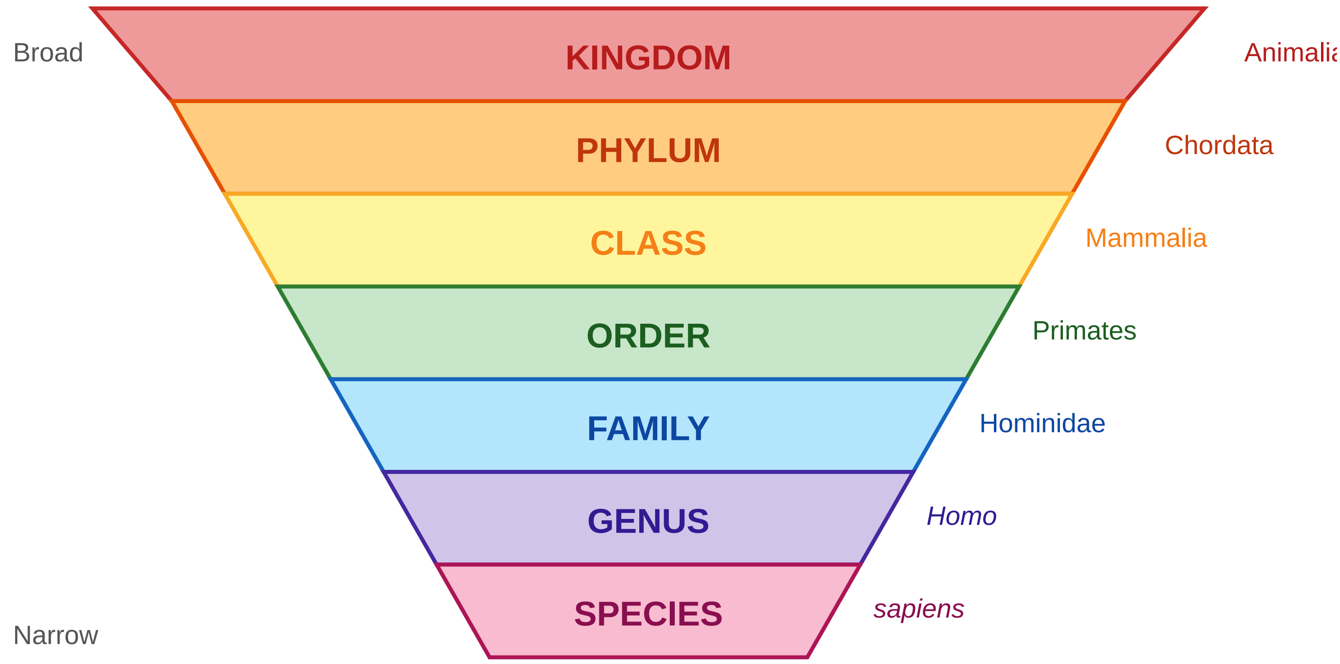
Chapter A1 — Classification of Life

The founder — Carl Linnaeus (1735)

- Swedish botanist; needed a universal filing system for every creature on Earth.
- Invented **binomial nomenclature** — two-part Latin names like *Homo sapiens*.
- Invented the **seven-rank hierarchy** still used today.
- **Exam tag:** "Father of Taxonomy" = Carl Linnaeus.

The seven taxonomic ranks

Kingdom → Phylum → Class → Order → Family → Genus → Species



Mnemonic: King Philip Came Over For Great Supper

REMEMBER

King Philip Came Over For Great Supper — Kingdom, Phylum, Class, Order, Family, Genus, Species. This mnemonic appears in textbooks from Class 6 to postgraduate level; it is absolutely exam-safe.

Whittaker's Five Kingdoms (1969)

Kingdom	Members	Key feature
Monera	Bacteria, cyanobacteria	Prokaryotes — no nucleus
Protista	Amoeba, Paramecium, Euglena	Unicellular eukaryotes
Fungi	Mushrooms, yeasts, moulds	Chitin cell wall; decomposers
Plantae	Mosses, ferns, flowering plants	Photosynthetic multicellular
Animalia	Insects, fish, mammals	Heterotrophic multicellular

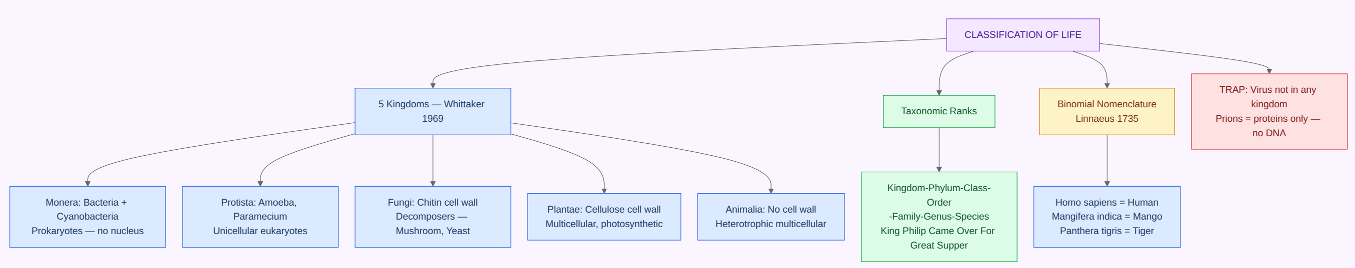
Later models added Archaea → six kingdoms. Carl Woese proposed three domains: Bacteria, Archaea, Eukarya.

KEY FACT

Exam-critical "Fathers" — memorise all five:
Father of taxonomy → **Carl Linnaeus**
Father of modern biology → **Aristotle**
Father of genetics → **Gregor Mendel**
Father of botany → **Theophrastus**

Father of zoology → **Aristotle** (same person)
Binomial name of humans → **Homo sapiens**

◎ **CHAPTER SUMMARY — EVERYTHING IN ONE VIEW**



Chapter A2 — The Cell

The discovery

- **1665** — Robert Hooke sliced cork under a primitive microscope.
- Saw small boxy chambers; called them **cells** (Latin *cellula* = "small room").
- **1838** — Schleiden formalised cell theory for plants.
- **1839** — Schwann formalised cell theory for animals.
- **1855** — Virchow added: "all cells arise from pre-existing cells."

Three tenets of Cell Theory

1. All living organisms are made of one or more cells.
2. The cell is the basic structural and functional unit of life.
3. New cells arise only from pre-existing cells. *(Virchow, 1855)*

Prokaryote vs Eukaryote — the fundamental split

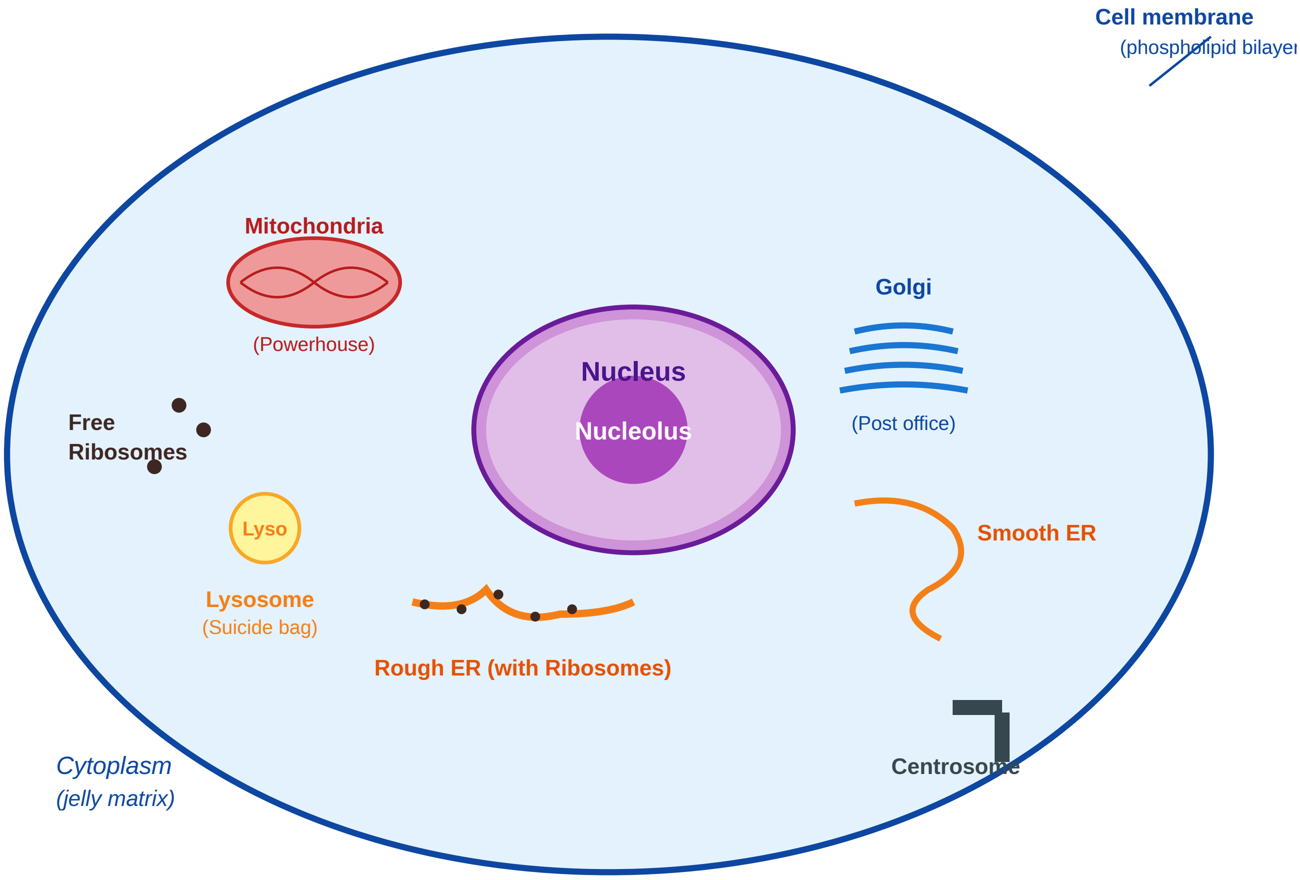
Feature	Prokaryote	Eukaryote
True nucleus	Absent (DNA in nucleoid)	Present
Size	1–10 µm	10–100 µm
Membrane-bound organelles	None	Many
Ribosomes	70S	80S (cytoplasm); 70S in mitochondria/chloroplast
Cell division	Binary fission	Mitosis / Meiosis
Examples	Bacteria, cyanobacteria	Plants, animals, fungi

Cell organelles — the factory analogy

Organelle	Function	Memory hook
Nucleus	DNA storage; controls cell	CEO's office
Mitochondria	ATP (energy) via respiration	**Power plant**
Ribosome	Protein synthesis	3-D printer
ER (rough)	Protein folding + transport	Assembly line
ER (smooth)	Lipid synthesis, detox	Pipeline
Golgi apparatus	Packages + dispatches proteins	Post office
Lysosome	Digests waste + worn-out parts	Recycling centre
Chloroplast	Photosynthesis (plants only)	Solar panel
Vacuole	Storage; turgor in plants	Warehouse
Cell wall	Rigidity; protection	Building walls
Centrosome	Spindle formation (animal cells)	Foreman
Peroxisome	Breaks H_2O_2 via catalase	Detox unit

EXAM ALERT

Four organelle questions examiners love:
"Powerhouse of the cell?" → **Mitochondria**
"Suicide bags?" → **Lysosomes** (contain hydrolytic enzymes; self-digest when cell dies)
"Site of protein synthesis?" → **Ribosomes**
"Which organelle has its own DNA?" → **Mitochondria AND Chloroplast** (both have own circular DNA)



Animal cell — typical eukaryote

Cell division — Mitosis vs Meiosis

Aspect	Mitosis	Meiosis
Location	Somatic (body) cells	Germ (reproductive) cells
Daughter cells	2	4
Chromosome count	Stays diploid (2n)	Halved to haploid (n)
Genetic variation	None — clones	High (crossing over)
Purpose	Growth, repair	Making gametes
Phases	PMAT × 1	PMAT × 2 (I + II)

REMEMBER

Phases of mitosis: PMAT — Prophase, Metaphase, Anaphase, Telophase. In meiosis, the same phases happen twice (Meiosis I and Meiosis II), but Prophase I has the special event of crossing over (chiasmata formation), which creates genetic variation.

Body extremes — memorise cold

Fact	Answer
Largest cell (single cell)	Ostrich egg

Fact	Answer
Smallest cell	*Mycoplasma*
Longest cell	Neuron (axon up to ~1 m)
First to see cells	Robert Hooke (1665)
Discovered nucleus	Robert Brown (1831)

PART B — GENETICS & EVOLUTION

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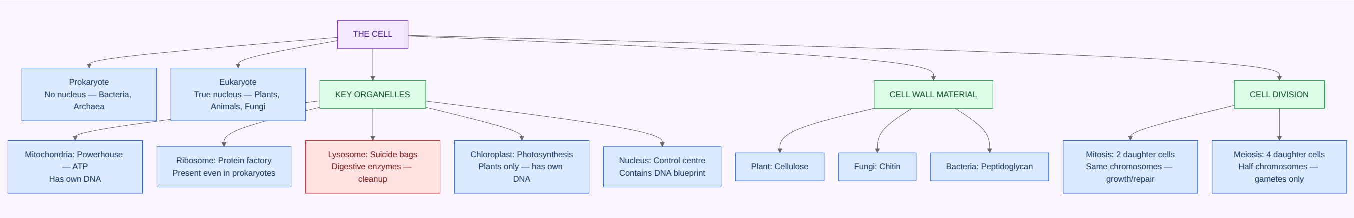

Chapter B1 — Mendelian Genetics

The founder — Gregor Mendel (1856–63)

- Augustinian friar in **Brno, Moravia**.
- Grew **28,000 pea plants** (*Pisum sativum*).
- Discovery: inheritance is **particulate**, not blended.
- Paper published **1866** — ignored for 34 years.
- **Rediscovered 1900** by De Vries, Correns, von Tschermak.
- **Exam tag:** "Father of Genetics."

Mendel's Three Laws

1. **Law of Dominance** — In a heterozygote (Tt), the dominant allele (T) masks the recessive allele (t). Only TT and Tt look tall; tt looks short.
2. **Law of Segregation** — Alleles separate during gamete formation. Each gamete gets only one allele from each pair.
3. **Law of Independent Assortment** — Alleles for different traits sort into gametes independently (provided genes are on different chromosomes).

Monohybrid cross: Tt × Tt

- **Phenotypic ratio → 3 : 1** (dominant : recessive)
- **Genotypic ratio → 1 TT : 2 Tt : 1 tt**

Punnett Square — Monohybrid Cross (Tt × Tt)

	T	t
T	**TT** Tall (pure)	**Tt** Tall (hybrid)
t	**Tt** Tall (hybrid)	**tt** Dwarf

Phenotype
3 Tall : 1 Dwarf

Genotype
1 TT : 2 Tt : 1 tt

Dihybrid (TtYy × TtYy)
→ 9 : 3 : 3 : 1
(2 traits at once)
Tested every paper.

Father gametes ↑ • Mother gametes ←

EXAM ALERT

Three questions that appear constantly:
"Father of genetics?" → **Gregor Mendel**
"Mendel used which plant?" → **Pea plant (Pisum sativum)**
"Classic monohybrid ratio?" → **3:1** (phenotypic) and **1:2:1** (genotypic)

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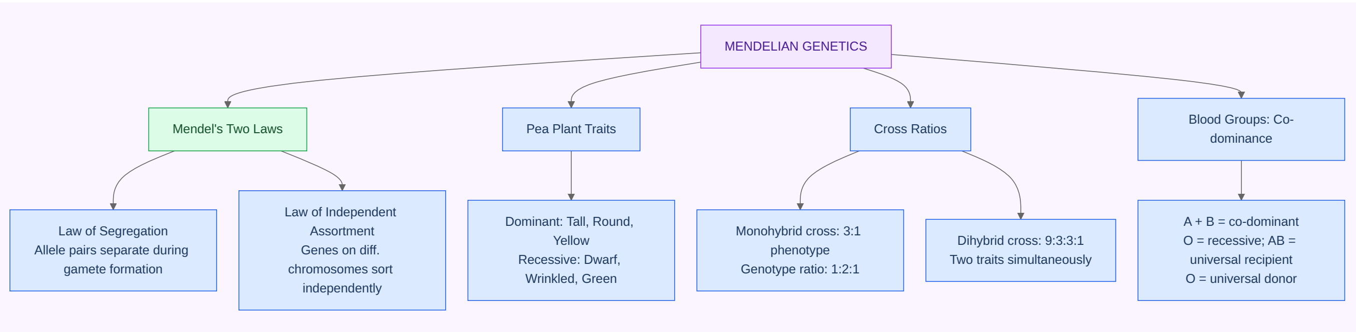


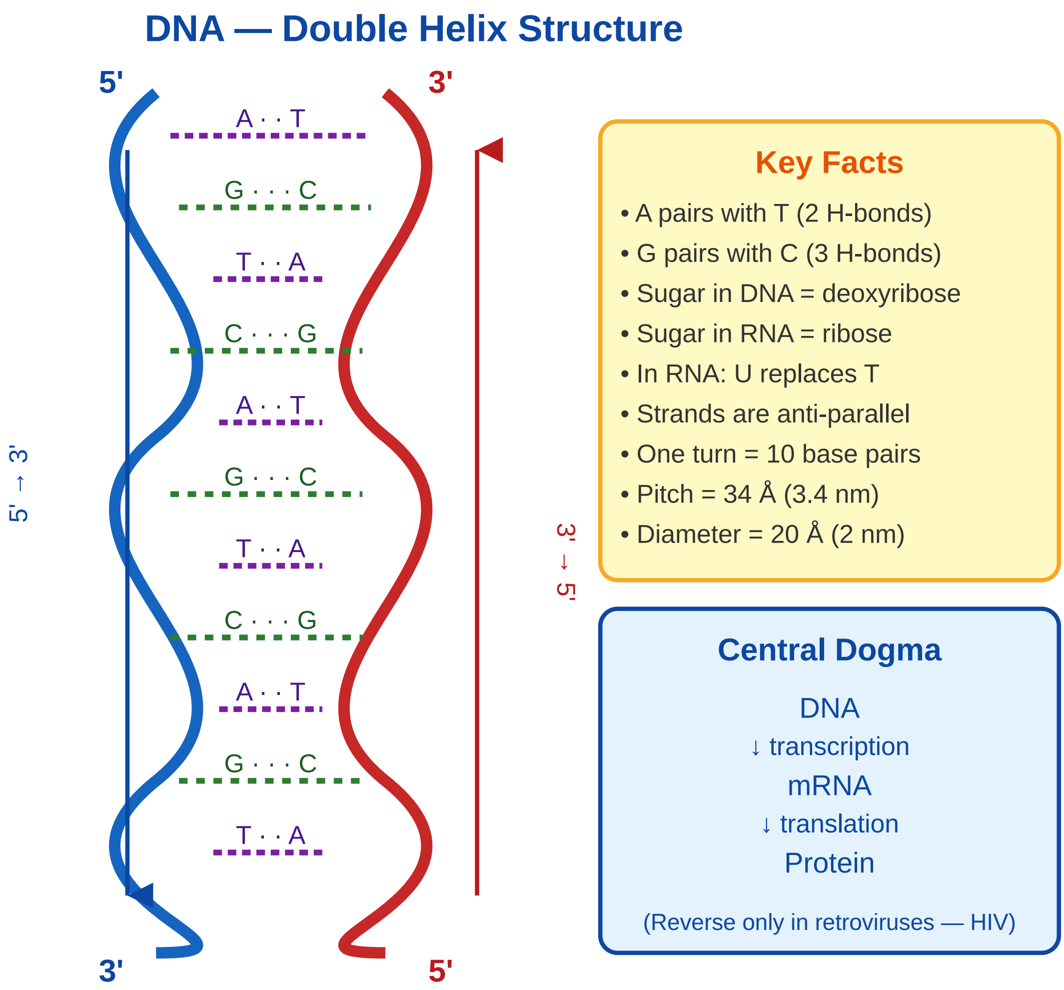
Chapter B2 — DNA & Molecular Genetics

The Double Helix

Discovery — 1953

- **James Watson + Francis Crick** built the model.
- Used X-ray diffraction data from **Rosalind Franklin** (Photo 51).

- **Maurice Wilkins** also contributed; shared the **Nobel Prize 1962**.
- Franklin had died (1958) and was not eligible posthumously.



Key structural facts: - Two strands run anti-parallel (5'→3' and 3'→5'). - Base pairing: **Adenine–Thymine** (2 hydrogen bonds), **Guanine–Cytosine** (3 hydrogen bonds). - Sugar in DNA = **deoxyribose**; in RNA = **ribose**. - Uracil (U) replaces Thymine (T) in RNA.

Central Dogma

DNA → (Transcription) → mRNA → (Translation) → Protein

Reverse transcription (RNA → DNA) occurs in retroviruses like HIV, via the enzyme reverse transcriptase.

Three types of RNA

RNA Type	Function
mRNA (messenger)	Carries genetic code from nucleus to ribosome
tRNA (transfer)	Brings amino acids to ribosome during translation
rRNA (ribosomal)	Structural component of ribosomes

KEY FACT

Genetic code essentials:

Start codon → **AUG** (codes for methionine)

Stop codons → **UAA, UAG, UGA**

Total codons → 64 (for 20 amino acids — degeneracy means multiple codons map to same amino acid)

Human chromosomes → **46 (23 pairs)**; 22 autosome pairs + 1 sex pair (XX female, XY male)

Human Genome Project → 1990–2003

CRISPR-Cas9 Nobel → Doudna + Charpentier (2020)

Genetic disorders

Disorder	Type	Key fact
Sickle cell anaemia	Autosomal recessive	Single mutation in haemoglobin β-chain
Thalassemia	Autosomal recessive	Reduced globin chain production
Haemophilia	X-linked recessive	Defective clotting factor (mainly males)
Colour blindness	X-linked recessive	X-linked; mostly males affected
Down syndrome	Trisomy 21	Extra chromosome 21
Turner syndrome	XO	Female with single X chromosome
Klinefelter syndrome	XXY	Male with extra X
Huntington's disease	Autosomal dominant	CAG repeat expansion; late onset

COMMON TRAP

Sex of the baby is determined by the **father's sperm**, not the mother. The mother always contributes X. Father contributes X → daughter; Y → son. Questions asking "who determines sex?" — the answer is the father.

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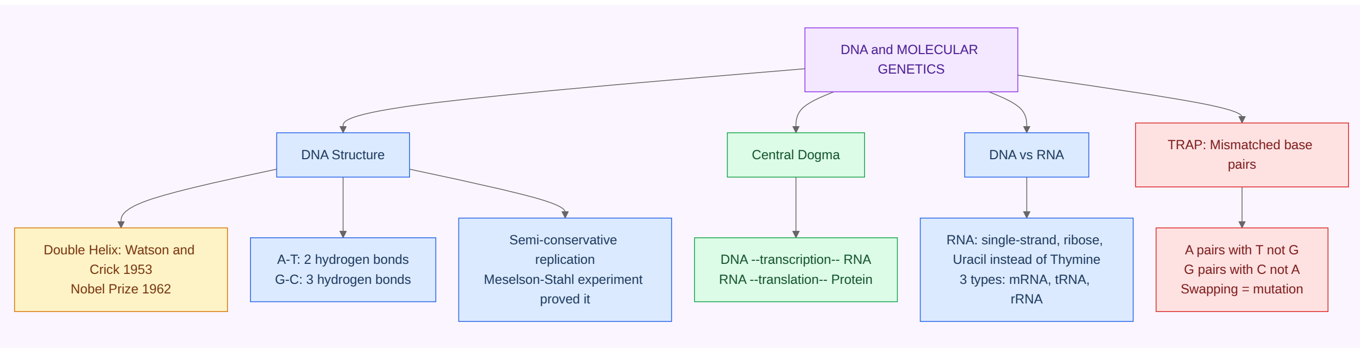


Chapter B3 — Evolution

Charles Darwin — the foundation

- **1831–36** — sailed on HMS Beagle around the world.

- Observed **Galápagos finches** with different beak shapes for different foods.
- **1859** — published *On the Origin of Species*.
- Proposed **natural selection** as the mechanism of evolution.
- Co-discovered independently by **Alfred Russel Wallace**.

Evidence for evolution

Evidence type	Example
Fossils	Archaeopteryx — link between birds and reptiles
Homologous organs	Human arm, whale flipper, bat wing — same structure, different function
Analogous organs	Insect wing, bird wing — same function, different origin
Vestigial organs	Appendix, coccyx, wisdom teeth
Molecular biology	DNA similarity across species
Comparative embryology	Vertebrate embryos look similar in early stages

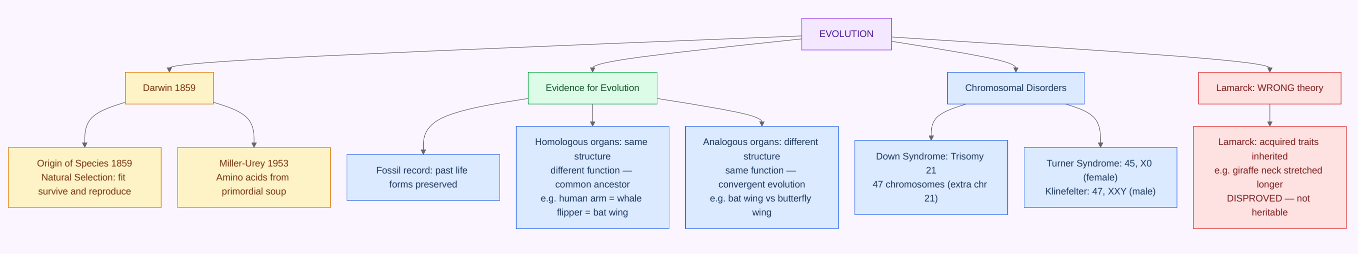
Human evolution: Australopithecus → Homo habilis → Homo erectus → Homo sapiens (~300,000 years ago, Africa).

KEY FACT

Key distinctions examiners test:
Homologous organs → same *origin*, different function (evolution divergence)
Analogous organs → same *function*, different origin (convergent evolution)
Lamarck's theory → inheritance of acquired characters (giraffe neck story) — **disproved**

PART C — PLANT BIOLOGY

◎ CHAPTER SUMMARY — EVERYTHING IN ONE VIEW



Chapter C1 — Plant Kingdom Overview

Plants are classified by:

- **Complexity of structure** — single cell → vascular tissue → flowering.
- **Reproductive strategy** — spores → naked seeds → enclosed seeds.

Division	Examples	Key feature
Thallophyta	Spirogyra, Chara, Algae	No roots/stems/leaves
Bryophyta	Mosses, Marchantia, Riccia	"Amphibians of plant kingdom"; no vascular tissue
Pteridophyta	Ferns, Equisetum, Selaginella	First vascular plants; no seeds
Gymnosperms	Pinus, Cycas, Cedrus	Naked seeds (no fruit)
Angiosperms	All flowering plants	Seeds enclosed in fruit

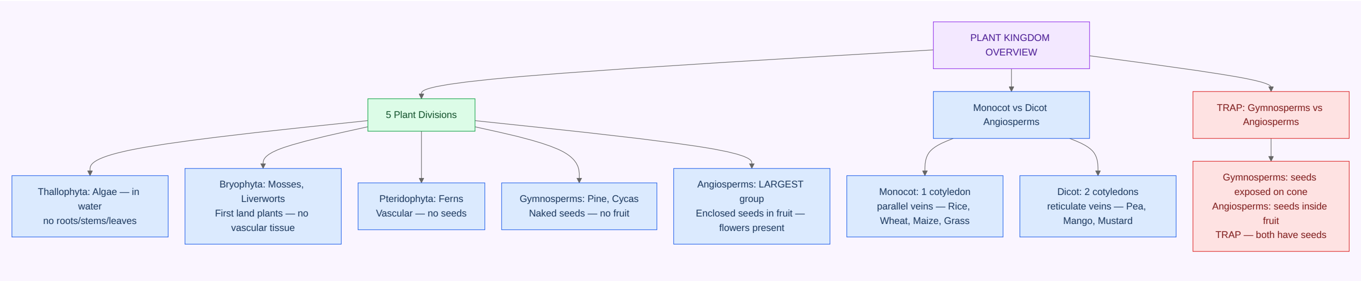
Dicot vs Monocot — always tested

Feature	Dicot	Monocot
Seed leaves (cotyledons)	2	1
Leaf venation	Reticulate (net-like)	Parallel
Flower parts	In 4s or 5s	In 3s
Root type	Tap root	Fibrous
Examples	Bean, mango, rose, pea	Wheat, rice, maize, onion, grass

REMEMBER

"One for monocots, two for dicots." All the world's staple grains (wheat, rice, maize, barley) are monocots. All the world's trees and most garden flowers are dicots. This pattern is an easy reality check.

◎ CHAPTER SUMMARY — EVERYTHING IN ONE VIEW



Chapter C2 — Photosynthesis

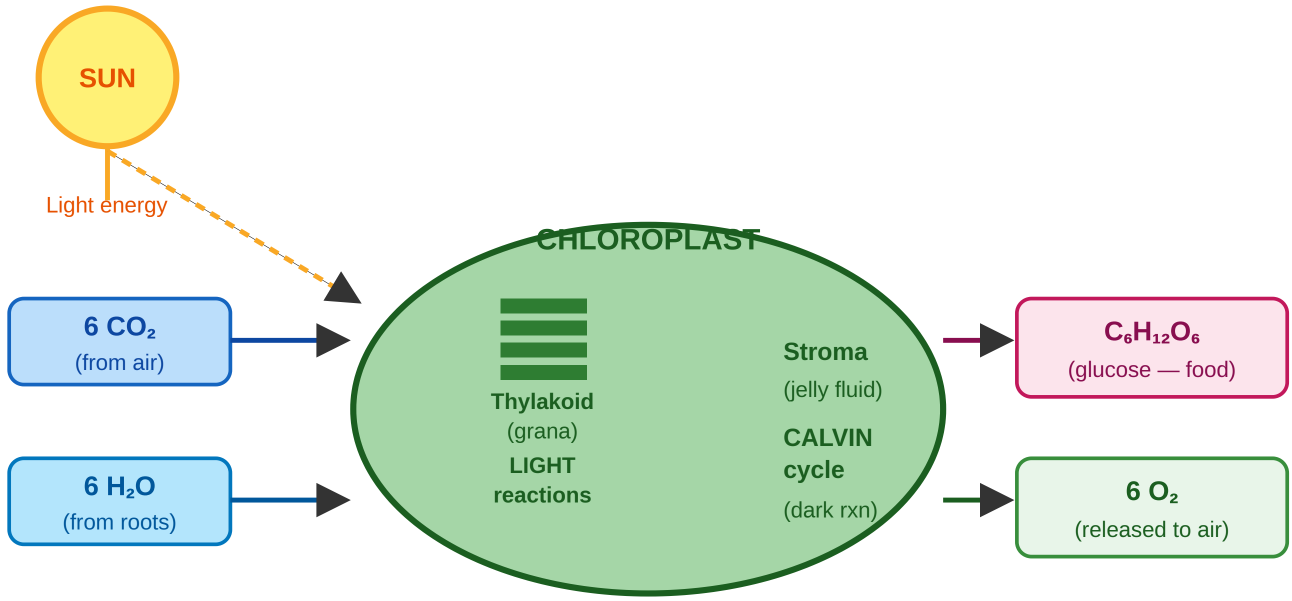
The big idea

- Plants manufacture food from **air + water** using **sunlight**.
- Site of action: **chloroplast** (specifically the thylakoid + stroma).
- Pigment that captures light: **chlorophyll-a** (primary).

The overall equation:

$6\ CO_2 + 6\ H_2O + \text{light} \rightarrow C_6H_{12}O_6 + 6\ O_2$

Photosynthesis — Inputs, Process, Outputs



Site of light reactions = thylakoid (grana) | Site of Calvin cycle = stroma | Chlorophyll-a is the primary pigment

Two phases

Light reaction (in thylakoids):

- Water is split (**photolysis**).
- **O_2** released as by-product.
- **ATP** and **NADPH** are produced.

Dark reaction / Calvin cycle (in stroma):

- ATP + NADPH consumed.
- **CO_2 fixed** into glucose.
- Called "dark" but actually runs during day too — uses products of the light reaction.

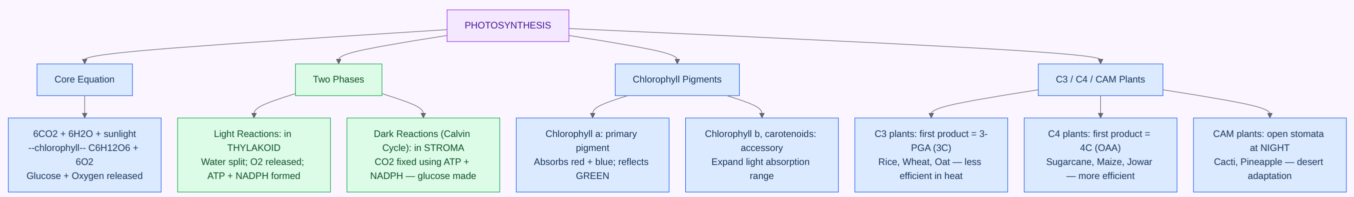
C3, C4, and CAM plants

Type	Plants	Advantage
C3	Wheat, rice, soybean	Most common; less efficient in heat
C4	Sugarcane, maize, sorghum	Efficient in tropical heat and high light
CAM	Cactus, pineapple, succulents	Open stomata at night; minimise water loss

EXAM ALERT

Top four photosynthesis questions:
"Site of photosynthesis?" → **Chloroplast**
"Main photosynthetic pigment?" → **Chlorophyll-a**
"Where does the Calvin cycle occur?" → **Stroma**
"CAM plants open stomata when?" → **Night** (to reduce water loss in desert conditions)

◎ CHAPTER SUMMARY — EVERYTHING IN ONE VIEW



Chapter C3 — Plant Hormones

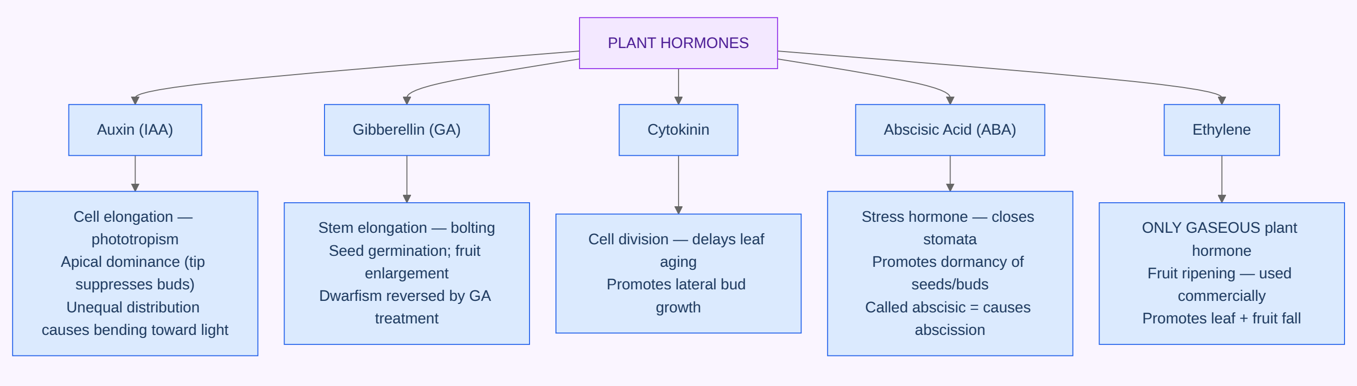
Hormone	Main function	Exam clue
Auxin (IAA)	Cell elongation, phototropism, apical dominance	Bends towards light
Gibberellin	Stem elongation, seed germination, breaks dormancy	Makes dwarf plants tall
Cytokinin	Cell division, delays leaf senescence	"Stay young" hormone
Abscisic acid (ABA)	Stress response, stomatal closure, seed dormancy	"Stress hormone"
Ethylene	Fruit ripening, leaf fall	The only gaseous plant hormone

KEY FACT

"Which hormone ripens fruit?" → **Ethylene** — it is a gas. Commercially, ethylene is used to ripen bananas and mangoes during transport.

"Hormone responsible for apical dominance?" → **Auxin** — suppresses lateral buds; pruning removes the apex and removes auxin dominance, causing bushy growth.

◎ CHAPTER SUMMARY — EVERYTHING IN ONE VIEW



Chapter C4 — Plant Reproduction

Asexual reproduction:

- Vegetative propagation — cutting, grafting, layering.
- Spore formation.
- Micropropagation (tissue culture).

Sexual reproduction:

- Pollination → Fertilisation → Seed.

Pollination agents (memorise the names):

Agent	Name	Example
Wind	Anemophily	Grass, maize, pine
Water	Hydrophily	Vallisneria
Insects	Entomophily	Rose, sunflower
Birds	Ornithophily	Hibiscus
Bats	Chiropterophily	Kigelia, baobab

Double fertilisation — unique to angiosperms:

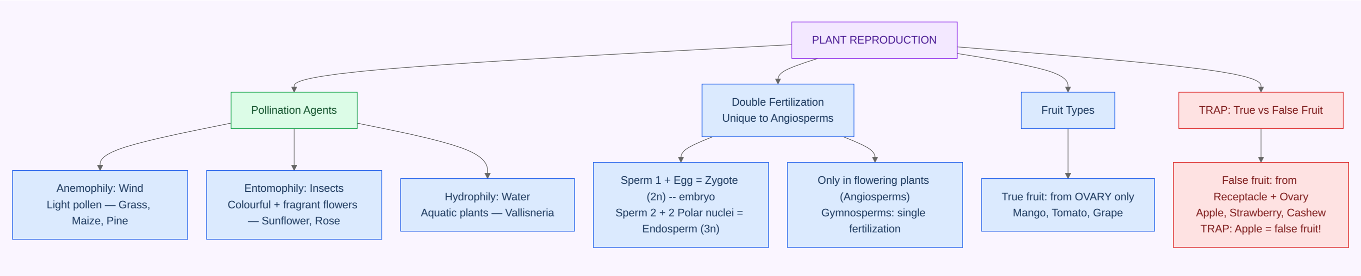
- **Sperm 1 + Egg** → Zygote (2n) → **embryo**.
- **Sperm 2 + 2 polar nuclei** → Endosperm (3n) → **food for embryo**.
- Only happens in flowering plants — gymnosperms have single fertilisation.

Famous Indian botanists

- **M.S. Swaminathan** — Father of the Indian Green Revolution (high-yield wheat varieties).
- **Birbal Sahni** — pioneer of palaeobotany in India.
- **J.C. Bose** — proved that plants respond to stimuli.

PART D — HUMAN BODY SYSTEMS

◎ CHAPTER SUMMARY — EVERYTHING IN ONE VIEW

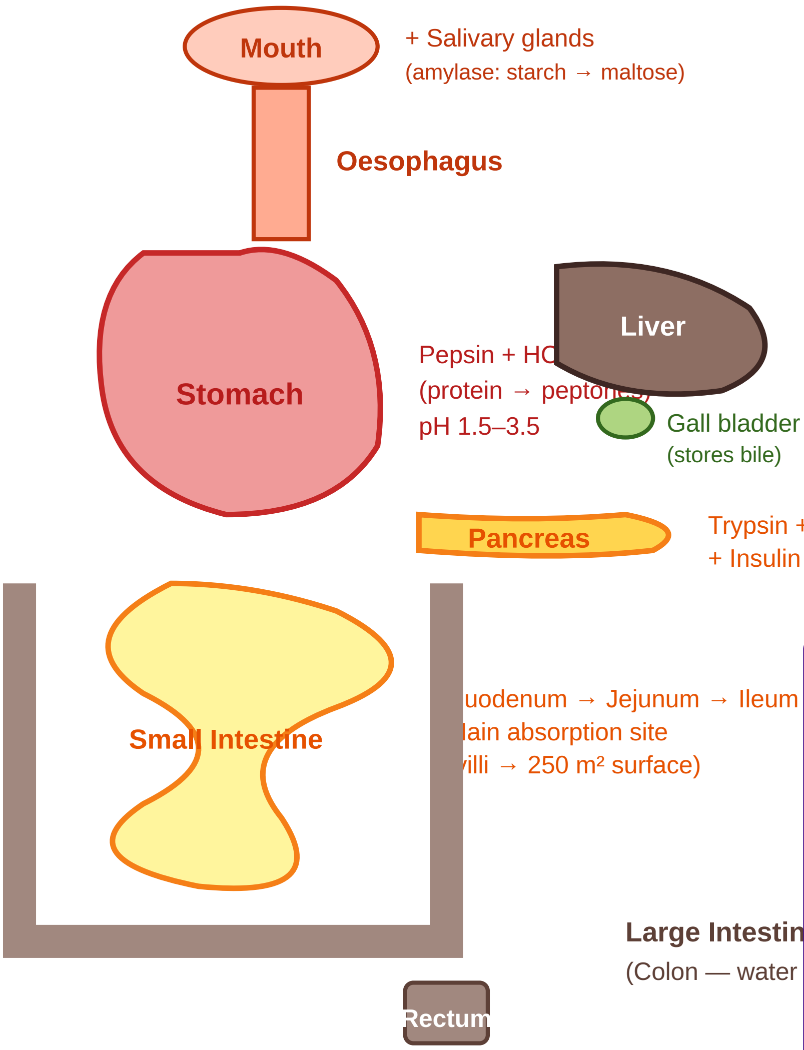


Chapter D1 — Digestive System

- The digestive tract runs about **9 metres** from mouth to anus.
- Each section specialises in a different stage of breakdown.

Pathway: Mouth → Pharynx → Oesophagus → Stomach → Small intestine (Duodenum → Jejunum → Ileum) → Large intestine (Caecum → Colon → Rectum) → Anus.

Digestive System — Pathway & Key Organs



Exam Quick-Hits

- Largest gland: Liver
- Bile stored: gall bladder
- Bile = NO enzyme (just emulsifies fat)
- Absorption: small intestine
- Water: large intestine
- Stomach pH: 1.5–3.5
- Total length: ~9 m
- Vitamin K made by gut bact.

Key enzymes

Source	Enzyme	Substrate → Product
Saliva	Amylase (Ptyalin)	Starch → Maltose
Stomach	Pepsin (+ HCl)	Proteins → Peptones
Pancreas	Amylase	Starch → Maltose
Pancreas	Lipase	Fats → Fatty acids + Glycerol
Pancreas	Trypsin, Chymotrypsin	Proteins → Peptides
Liver (via bile)	—	Emulsifies fats
Small intestine	Maltase, Lactase, Sucrase	Disaccharides → Monosaccharides

KEY FACT

Three digestive anatomy questions examiners love:
"Largest gland in the body?" → **Liver**
"Bile is stored in?" → **Gall bladder**
"Main site of nutrient absorption?" → **Small intestine** (villi and microvilli maximise surface area to ~250 m²)
"Stomach acid pH?" → **1.5–3.5**

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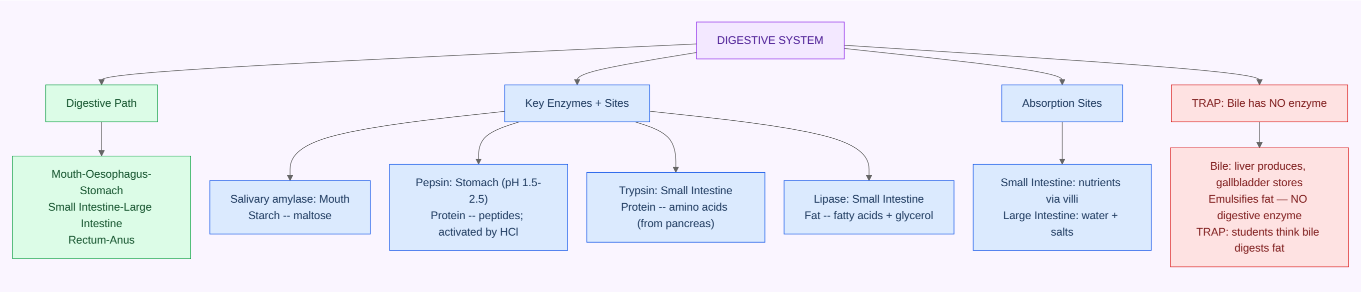

Chapter D2 — Respiratory System

Pathway: Nose → Pharynx → Larynx → Trachea → Bronchi → Bronchioles → Alveoli.

- ~300 million alveoli; surface area ~70 m².
- Diaphragm and intercostal muscles drive breathing.
- Normal adult breathing rate: **12–18 per minute**; tidal volume ~500 mL.
- O_2 transported mainly by **haemoglobin** (Hb); CO_2 transported mainly as **bicarbonate (HCO_3^-)** (~70%).

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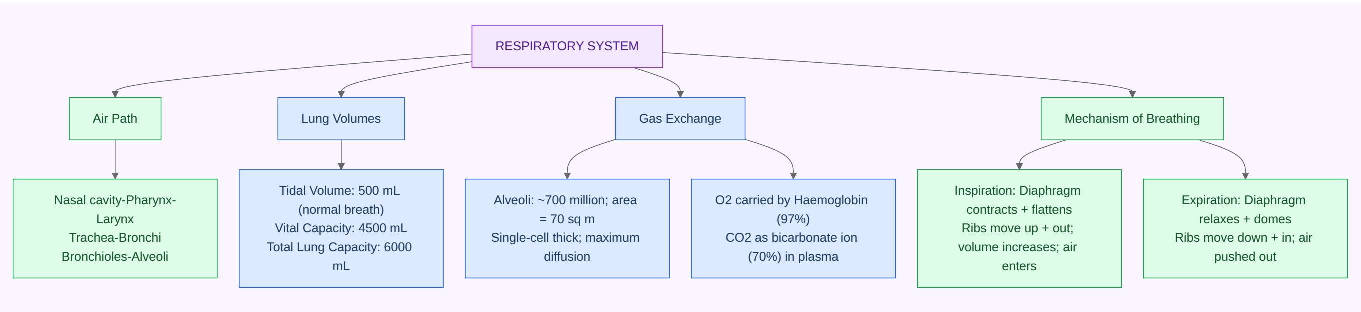
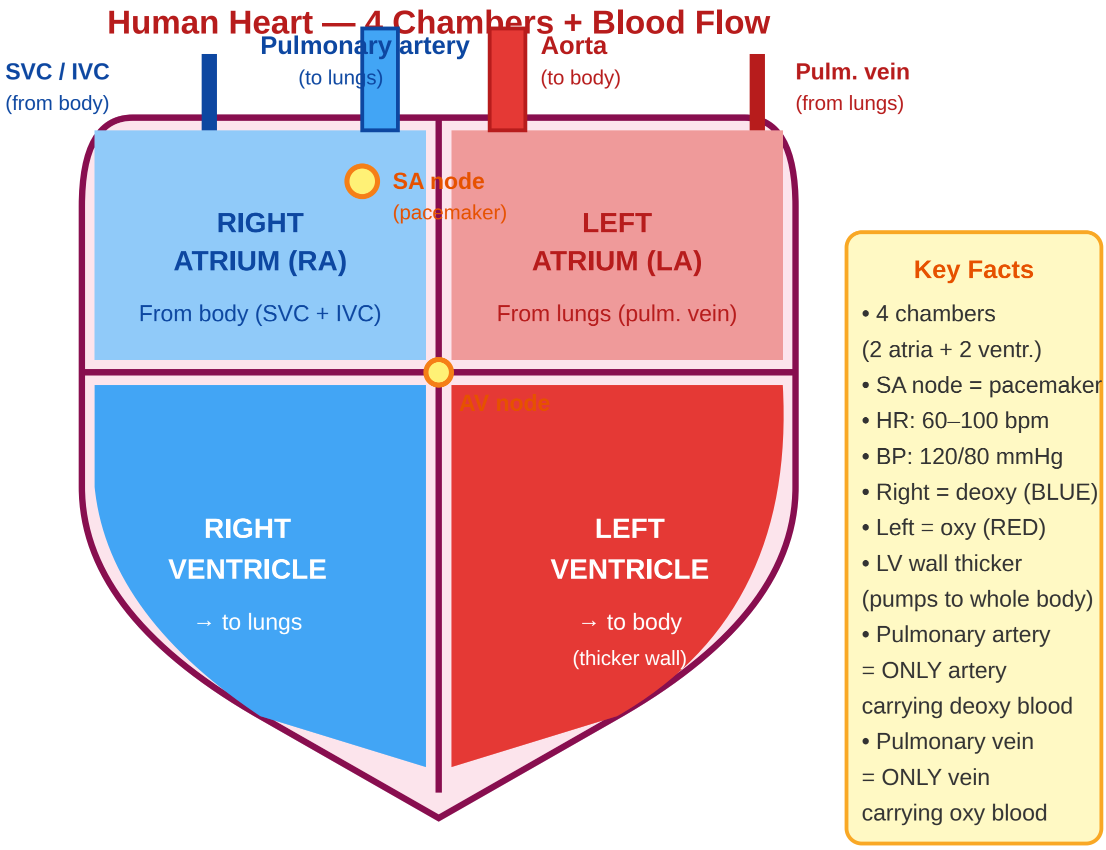


Chapter D3 — Circulatory System

The heart

- **4 chambers** — 2 atria (upper) + 2 ventricles (lower).
- **Right heart** — receives deoxygenated blood → pumps to lungs.
- **Left heart** — receives oxygenated blood from lungs → pumps to body.
- **Natural pacemaker** — SA node (sinoatrial node) in right atrium.
- **Normal heart rate** — 60–100 bpm.

- **Normal blood pressure — 120 / 80 mmHg** (systolic / diastolic).



→ Body → RA → RV → Lungs → LA → LV → Body (double circulation, Harvey 1628)

Blood composition

Component	Amount	Function
Plasma	~55% of blood	Transport medium; contains proteins, hormones
RBCs	~45%; 4–6 million/µL	O_2 transport via haemoglobin; no nucleus; lifespan 120 days
WBCs	4,000–11,000/µL	Immune defence
Platelets	150,000–450,000/µL	Clotting

EXAM ALERT

Blood group questions — 100% exam reliable:
"Universal donor?" → **O negative (O−)** — no antigens on RBCs, no Rh factor
"Universal recipient?" → **AB positive (AB+)** — no anti-A or anti-B antibodies in plasma
"Who discovered blood groups?" → **Karl Landsteiner** (Nobel Prize 1930)
"Normal Hb: male/female?" → **13–17 / 12–15.5 g/dL**

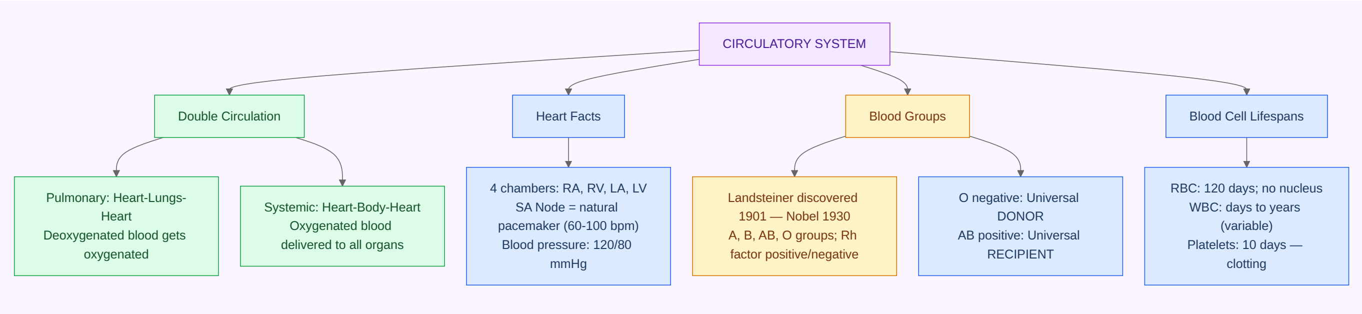
WBC types (granulocytes + agranulocytes)

WBC	%	Function
Neutrophils	60–70%	First responders to bacterial infection
Lymphocytes	20–40%	T cells (cellular immunity) + B cells (antibodies)
Monocytes	3–8%	Largest WBC; become macrophages
Eosinophils	1–4%	Parasites and allergic reactions
Basophils	<1%	Histamine; smallest %

COMMON TRAP

Examiners trick students with: "Biggest WBC?" → **Monocyte**. "Smallest WBC?" → **Lymphocyte**. Many students confuse this with platelets, which are not WBCs at all.

◎ CHAPTER SUMMARY — EVERYTHING IN ONE VIEW



Chapter D4 — Nervous System

Divisions

- **Central Nervous System (CNS)**: Brain + Spinal cord.
- **Peripheral Nervous System (PNS)**: All nerves outside CNS.
- Somatic: voluntary muscles.
- Autonomic: involuntary (sympathetic = fight/flight; parasympathetic = rest/digest).

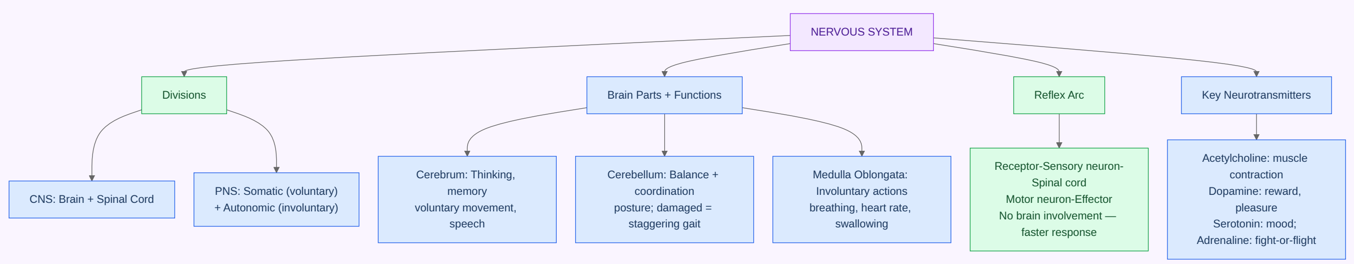
Brain regions

Region	Function
Cerebrum	Largest part; thinking, memory, voluntary action, speech
Cerebellum	Balance, coordination, posture
Medulla oblongata	Controls breathing, heart rate, blood pressure
Hypothalamus	Hunger, thirst, temperature regulation, sleep
Thalamus	Sensory relay station

KEY FACT

Nervous system exam essentials:
"Largest part of brain?" → **Cerebrum**
"Balance centre?" → **Cerebellum**
"Hunger and thirst centre?" → **Hypothalamus**
"Longest cell in the human body?" → **Neuron**
"Speed of nerve impulse in myelinated fibre?" → **~120 m/s**
Key neurotransmitters: acetylcholine, dopamine, serotonin, GABA, noradrenaline

◎ CHAPTER SUMMARY — EVERYTHING IN ONE VIEW



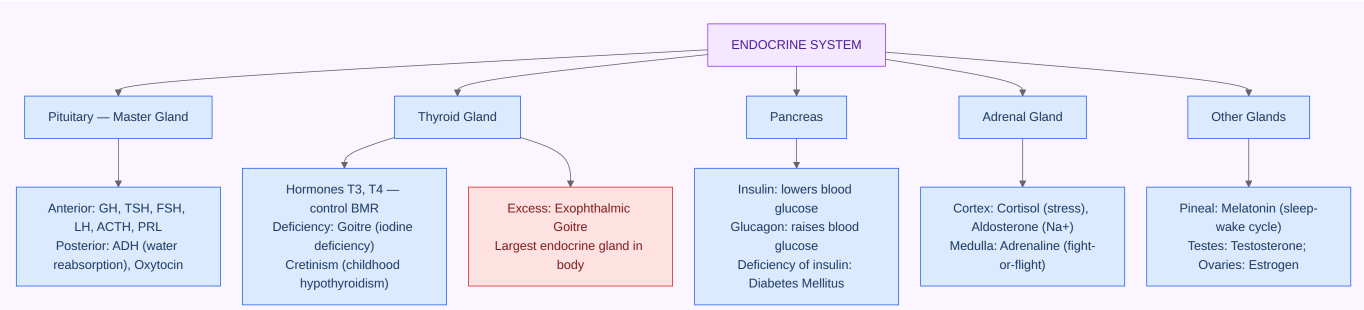
Chapter D5 — Endocrine System

Gland	Hormone(s)	Function	Disorder
Pituitary (anterior)	GH, TSH, ACTH, FSH, LH, Prolactin	"Master gland" — controls other glands	GH excess → Gigantism (child), Acromegaly (adult)
Pituitary (posterior)	ADH (Vasopressin), Oxytocin	Water balance; uterine contraction	ADH deficiency → Diabetes insipidus
Thyroid	T3, T4, Calcitonin	Metabolism; Ca regulation	Goitre (iodine deficiency); Grave's (hyper); Cretinism (hypo, child)
Parathyroid	PTH	Raises blood calcium	Tetany (deficiency)
Adrenal medulla	Adrenaline, Noradrenaline	Fight-or-flight response	—
Adrenal cortex	Cortisol, Aldosterone	Stress; Na+/K+ balance	Cushing's (excess); Addison's (deficiency)
Pancreas (β-cells)	Insulin	Lowers blood glucose	Diabetes mellitus Type 1 and 2
Pancreas (α-cells)	Glucagon	Raises blood glucose	—
Pineal	Melatonin	Sleep-wake cycle	—
Thymus	Thymosin	T-cell maturation	—

EXAM ALERT

Top endocrine exam questions:
"Master gland?" → **Pituitary**
"Goitre caused by?" → **Iodine deficiency → enlarged thyroid**
"Largest endocrine gland?" → **Thyroid**
"Fight-or-flight hormone?" → **Adrenaline (Epinephrine)**
"Insulin deficiency → ?" → **Diabetes mellitus**
"Love hormone / milk ejection?" → **Oxytocin**

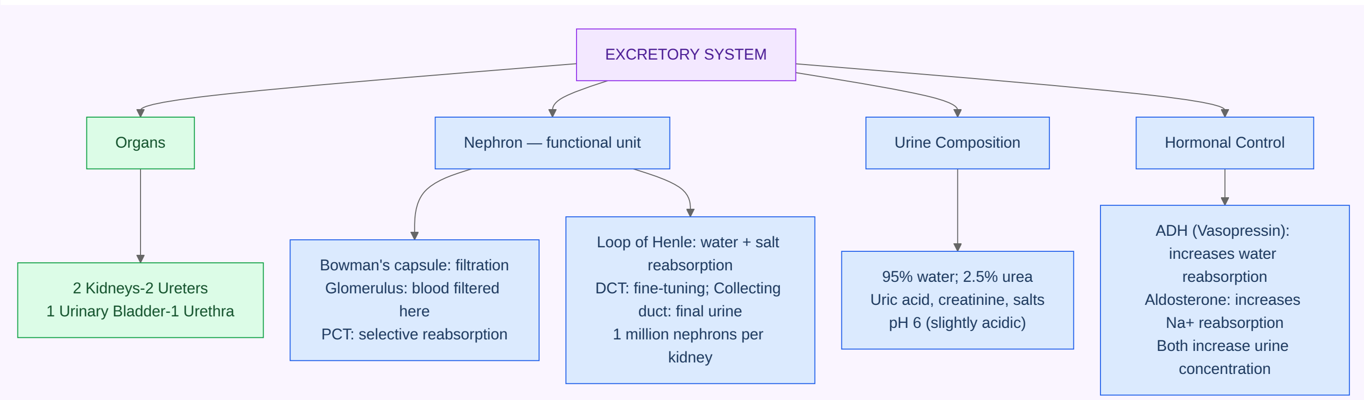
◎ CHAPTER SUMMARY — EVERYTHING IN ONE VIEW



Chapter D6 — Excretory System

- **Kidneys** (2): each contains ~1 million nephrons.
- **Nephron**: functional unit — glomerulus filters blood; Bowman's capsule collects filtrate; tubules reabsorb glucose, water, salts; Loop of Henle concentrates urine.
- Urine: ~96% water + urea + uric acid + creatinine + salts. Yellow colour from **urochrome**.
- Normal output: ~1.5 L/day.
- Other excretory organs: lungs (CO_2), skin (salts + water via sweat), liver (bile pigments).

◎ CHAPTER SUMMARY — EVERYTHING IN ONE VIEW



Chapter D7 — Skeletal & Muscular System

Skeleton

- Adult human: **206 bones** (infant has ~270; many fuse during development).

- **Largest bone**: Femur (thigh).
- **Smallest bone**: Stapes (middle ear).
- **Hardest substance in body**: Tooth enamel.
- Vertebral column: 33 vertebrae → fused to 26 in adults (7 cervical + 12 thoracic + 5 lumbar + sacrum + coccyx).

Joints

- Ball-and-socket: hip, shoulder (360° movement).
- Hinge: knee, elbow (one plane).
- Pivot: atlas–axis joint (rotation of head).

Muscles

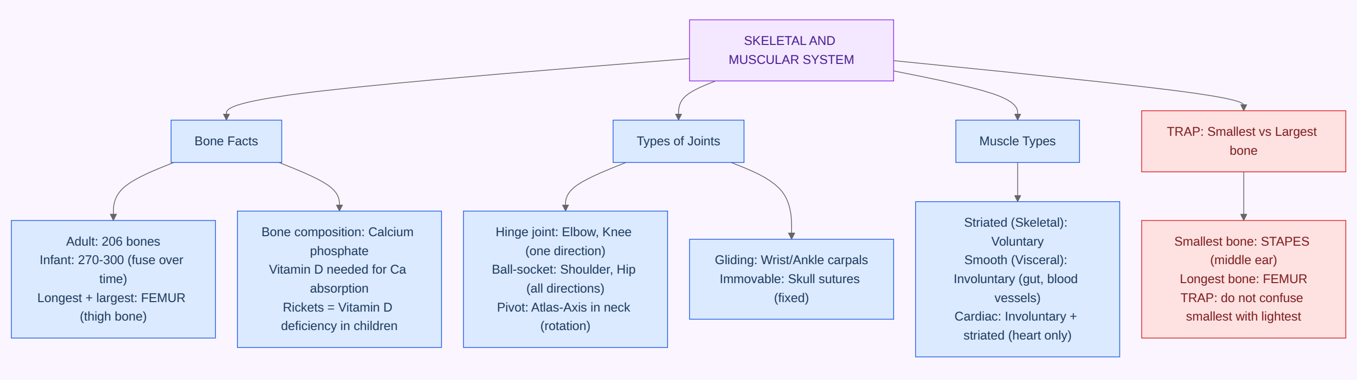
Type	Control	Structure	Example
Skeletal	Voluntary	Striated	Biceps, quadriceps
Smooth	Involuntary	Non-striated	Walls of gut, blood vessels
Cardiac	Involuntary	Striated	Heart wall — unique combination

KEY FACT

Key muscle and bone facts:
"Largest muscle?" → **Gluteus maximus**
"Smallest muscle?" → **Stapedius** (controls stapes in ear)
"Strongest muscle (force/area)?" → **Masseter** (jaw)
"Cardiac muscle is unique because?" → It is **involuntary AND striated** — no other muscle is both.

PART E — NUTRITION: VITAMINS & MINERALS

◎ CHAPTER SUMMARY — EVERYTHING IN ONE VIEW



Chapter E1 — Vitamins (Complete Table)

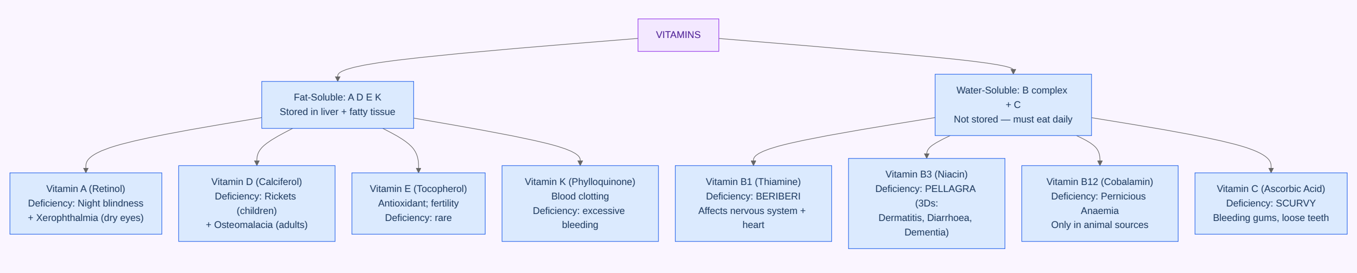
Vitamin	Chemical name	Sources	Deficiency disease
A	Retinol / β-carotene	Carrot, liver, milk	**Night blindness**, xerophthalmia
B_1	Thiamine	Whole grains, pulses	**Beri-beri**
B_2	Riboflavin	Milk, eggs	Cheilosis, ariboflavinosis
B_3	Niacin	Meat, peanuts	**Pellagra** (3 Ds: Dermatitis, Diarrhoea, Dementia)
B_5	Pantothenic acid	Widely present	Burning feet syndrome
B_6	Pyridoxine	Fish, poultry, banana	Anaemia
B_7	Biotin	Egg yolk, nuts	Hair loss, skin rash
B_9	Folic acid	Leafy greens	**Megaloblastic anaemia**; neural tube defects in foetus
B_{12}	Cyanocobalamin	Meat, fish, dairy (animal sources only)	**Pernicious anaemia**
C	Ascorbic acid	Amla, citrus, guava	**Scurvy** (bleeding gums, weak capillaries)
D	Calciferol	Sunlight, fish liver oil	**Rickets** (children), **Osteomalacia** (adults)
E	Tocopherol	Wheat germ, vegetable oils	Sterility
K	Phylloquinone	Leafy greens, gut bacteria	**Excessive bleeding** (clotting failure)

EXAM ALERT

The vitamin exam cheat sheet — learn these cold:
Fat-soluble → **A, D, E, K** (stored in body fat; can accumulate to toxic levels)

Water-soluble → **B-complex + C** (excreted in urine; need daily intake)
Only vitamin NOT found in plants → **B_{12}** (vegans must supplement)
Vitamin made by skin + sunlight → **D**
Blood clotting vitamin → **K**
Scurvy → **C** | Night blindness → **A** | Rickets → **D** | Beri-beri → **B_1**

◎ CHAPTER SUMMARY — EVERYTHING IN ONE VIEW

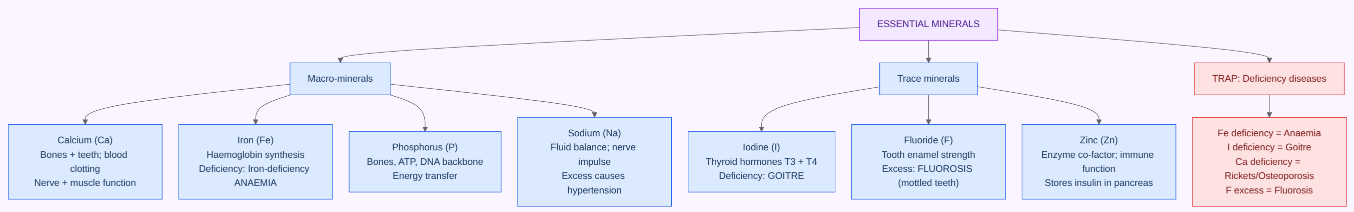


Chapter E2 — Essential Minerals

Mineral	Function	Deficiency
Iron	Haemoglobin synthesis	**Anaemia**
Iodine	Thyroid hormone synthesis	**Goitre**
Calcium	Bones, teeth, muscle contraction	Rickets, osteoporosis
Phosphorus	Bones, ATP, DNA	Rare
Sodium	Fluid balance, nerve impulse	Hyponatraemia
Potassium	Nerve and muscle function	Hypokalaemia
Fluorine	Dental enamel hardening	Dental caries (deficiency); Fluorosis (excess)
Zinc	Enzyme cofactor, immunity	Poor wound healing, immune deficiency

PART F — DISEASES & IMMUNITY

◎ CHAPTER SUMMARY — EVERYTHING IN ONE VIEW



Chapter F1 — Disease Classification

Every exam disease question falls into one of four buckets:

- **Pathogen type** — bacterial / viral / protozoal / fungal.
- **Vector** — Anopheles (malaria), Aedes (dengue), Culex (filaria).
- **Vaccine** — BCG (TB), DPT, MMR, OPV / IPV, COVID vaccines.
- **Elimination status** — smallpox (1980), polio in India (2014).

Bacterial diseases

Disease	Causative agent	Vaccine/treatment
Tuberculosis (TB)	*Mycobacterium tuberculosis*	**BCG** vaccine
Cholera	*Vibrio cholerae*	ORS; vaccine
Typhoid	*Salmonella typhi*	TAB vaccine
Plague	*Yersinia pestis*	Antibiotics; rat-flea vector
Tetanus	*Clostridium tetani*	**DPT** vaccine
Diphtheria	*Corynebacterium diphtheriae*	**DPT** vaccine
Whooping cough	*Bordetella pertussis*	**DPT** vaccine
Leprosy	*Mycobacterium leprae*	MDT (multi-drug therapy)

Viral diseases

Disease	Virus	Notes
Smallpox	Variola	**Eradicated 1980** — first human disease ever eradicated
Polio	Poliovirus	India polio-free since **27 March 2014**; OPV (Sabin) + IPV (Salk)
Measles	Paramyxovirus	MMR vaccine
AIDS	HIV (retrovirus)	First reported 1981; reverse transcriptase
Dengue	DENV 1–4	Vector: **female Aedes aegypti** (day-biter)

Disease	Virus	Notes
Rabies	Rabies lyssavirus	Post-exposure vaccine; dog/bat bite
COVID-19	SARS-CoV-2	Pandemic declared 11 March 2020
Hepatitis B	HBV	Blood/sexual transmission; vaccine available

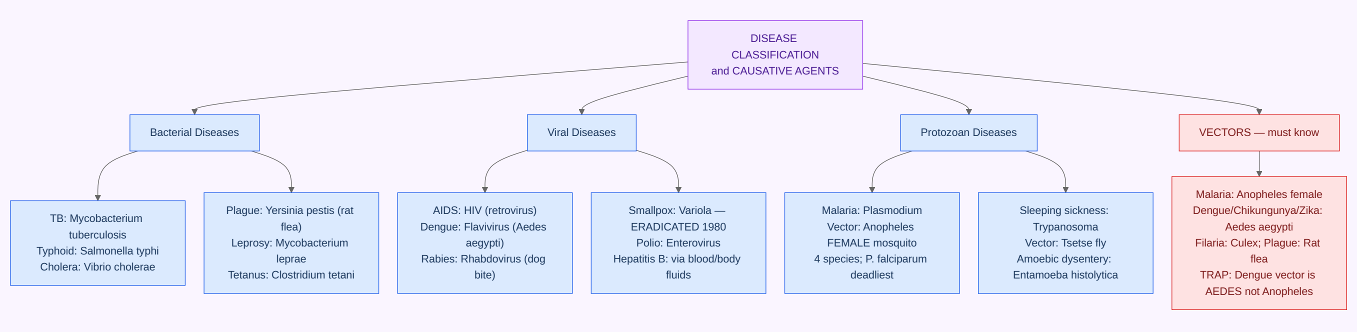
Protozoal & parasitic diseases

Disease	Pathogen	Vector
Malaria	*Plasmodium* spp.	**Female Anopheles** mosquito
Kala-azar	*Leishmania donovani*	Sand fly
Sleeping sickness	*Trypanosoma*	Tsetse fly
Filariasis	*Wuchereria bancrofti*	**Culex** mosquito
Amoebiasis	*Entamoeba histolytica*	Faecal-oral
Tapeworm	*Taenia* spp.	Undercooked pork/beef

EXAM ALERT

Vector mapping — tested in EVERY paper:
Malaria → Female **Anopheles**
Dengue → Female **Aedes aegypti** (also Zika, Chikungunya)
Filariasis → **Culex**
Kala-azar → **Sand fly**
Plague → **Rat flea**
Sleeping sickness → **Tsetse fly**

◎ CHAPTER SUMMARY — EVERYTHING IN ONE VIEW



Chapter F2 — Immunity

The immune system has two arms. **Innate immunity** is the rapid, non-specific response: skin as a barrier, mucus, and phagocytes that eat any invader without needing prior exposure. **Adaptive immunity** is specific and has memory: B lymphocytes produce antibodies; T lymphocytes kill infected cells directly.

Antibodies (Immunoglobulins)

Class	Key fact
IgG	Most abundant; crosses placenta; main antibody of secondary response
IgM	**First responder** — appears first after infection
IgA	Found in saliva, tears, breast milk (mucosal immunity)
IgE	Allergic reactions; parasites
IgD	B-cell activation

Vaccines

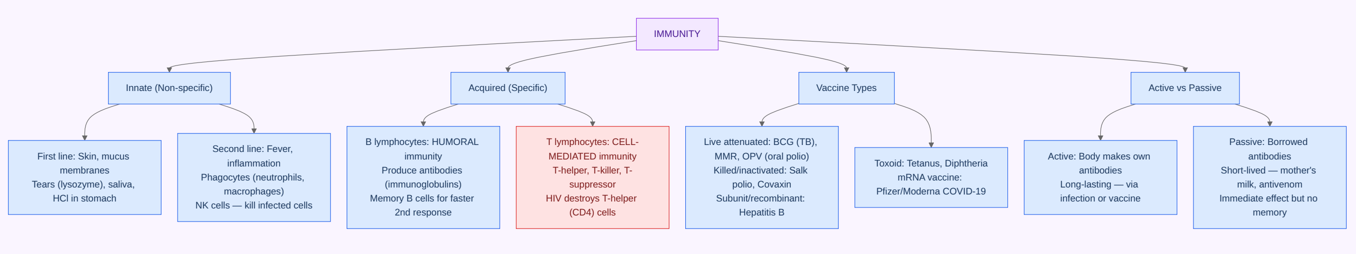
Vaccine	Disease
BCG	Tuberculosis
OPV (Sabin) / IPV (Salk)	Polio
MMR	Measles, Mumps, Rubella
DPT	Diphtheria, Pertussis, Tetanus
Hepatitis B	HBV
HPV	Human papillomavirus (prevents cervical cancer)

KEY FACT

Vaccine history points examiners test:
Concept of vaccination pioneered by → **Edward Jenner** (smallpox, 1796, using cowpox)
"First-responder antibody?" → **IgM**
"Most abundant antibody?" → **IgG**
"BCG stands for?" → Bacillus Calmette-Guérin
India's UIP (Universal Immunisation Programme) started → **1985**

PART G — ECOLOGY & ENVIRONMENT

◎ CHAPTER SUMMARY — EVERYTHING IN ONE VIEW



Chapter G1 — Ecosystem & Energy Flow

What is an ecosystem?

- A community of **living organisms (biotic)** interacting with the **non-living environment (abiotic)**.
- Operates in a defined area.
- Can be tiny (a pond, a tree) or huge (a forest, the entire **biosphere**).

Energy flow rules

- Flow is **one-directional**: Sun → Producers → Consumers → Decomposers. Energy cannot flow backwards.
- **10% rule (Lindeman's law)**: Only 10% of energy at one trophic level transfers to the next. 90% is lost as heat. This limits food chains to 4–5 links.
- **Trophic levels**: T1 = Producers (plants); T2 = Herbivores; T3 = Carnivores; T4 = Top predators. Decomposers operate at all levels.

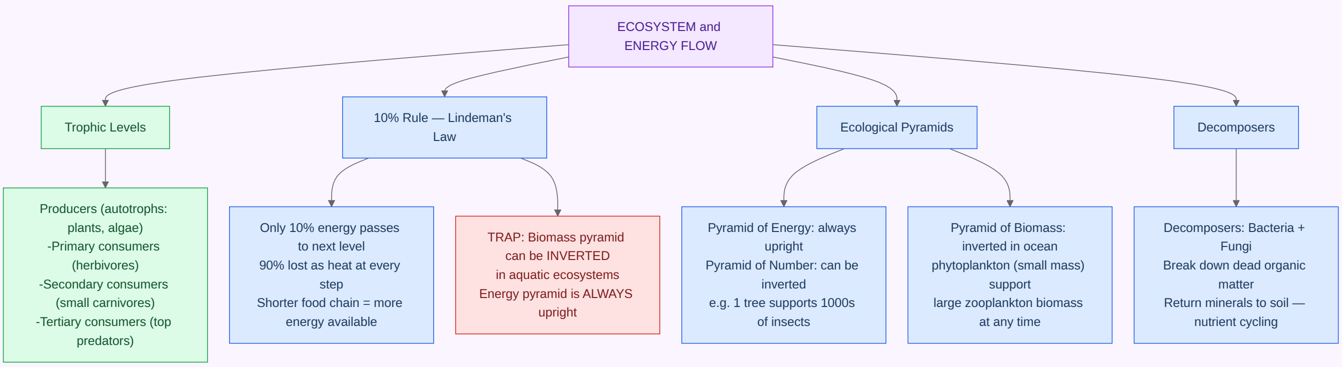
Ecological pyramids

Pyramid type	Always upright?	Why can it invert?
Numbers	No	One tree supports millions of insects
Biomass	No	Ocean: tiny phytoplankton support large fish biomass above
Energy	**YES — always**	Energy always decreases at each level; never inverts

KEY FACT

The pyramid of **energy** is ALWAYS upright — this is the most tested ecological fact. The pyramids of numbers and biomass can invert (inverted pyramids are possible). If the question says "which pyramid is always upright?" — the answer is Energy.

◎ CHAPTER SUMMARY — EVERYTHING IN ONE VIEW



Chapter G2 — Biodiversity

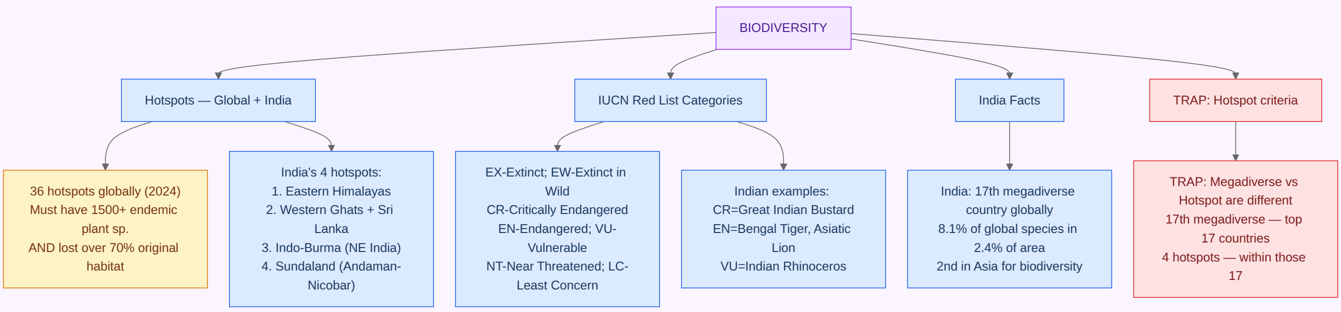
Three levels of biodiversity: Genetic diversity, Species diversity, Ecosystem diversity.

Biodiversity hotspots (globally 36; defined by >1,500 endemic plants AND >70% habitat lost):

India has **4 hotspots**: 1. Western Ghats (+ Sri Lanka) 2. Eastern Himalayas (Indo-Burma) 3. Indo-Burma (part overlapping NE India) 4. Sundaland (Andaman & Nicobar Islands)

IUCN Red List categories (most to least threatened): Extinct (EX) → Extinct in Wild (EW) → Critically Endangered (CR) → Endangered (EN) → Vulnerable (VU) → Near Threatened (NT) → Least Concern (LC)

◎ CHAPTER SUMMARY — EVERYTHING IN ONE VIEW



Chapter G3 — Indian Wildlife & Conservation

Major conservation projects

Project	Year	Target
Project Tiger	1973	Bengal tiger; 55+ reserves
Project Elephant	1992	Asian elephant
Project Crocodile	1975	Mugger, saltwater, gharial
Cheetah reintroduction	2022	Kuno National Park (Namibian + South African cheetahs)
Sea Turtle Project	1976	Olive ridley turtles

National symbols

- National animal: **Bengal tiger** (since 1973).
- National bird: **Indian peacock** (*Pavo cristatus*).
- National aquatic animal: **Ganges river dolphin** (since 2009).
- National heritage animal: **Elephant** (since 2010).
- National tree: **Indian banyan** (*Ficus benghalensis*).
- National flower: **Lotus** (*Nelumbo nucifera*).

Protected areas in India

- **National Parks**: ~106 (Jim Corbett was first, established 1936 as Hailey NP).
- **Wildlife Sanctuaries**: ~570.
- **Biosphere Reserves**: 18 (12 under UNESCO MAB programme).
- **Largest National Park**: Hemis (Ladakh).
- **Ramsar Sites** (wetlands): ~85 as of 2026.

EXAM ALERT

Conservation questions — high hit rate:
"India's first National Park?" → **Jim Corbett** (established 1936 as Hailey NP)
"Largest NP in India?" → **Hemis (Ladakh)**
"CITES regulates?" → International trade in endangered species
"Biodiversity Act in India?" → **2002** (follows CBD 1992 Rio)

APPENDIX A — Human Body Trivia (Memorise Cold)

Fact	Value
Bones (adult)	**206**
Bones (infant)	~270
Muscles	~640
Teeth (adult)	32
Teeth (milk/deciduous)	20
Largest bone	Femur
Smallest bone	Stapes (middle ear)
Largest organ	Skin
Largest internal organ	Liver
Largest gland	Liver
Master gland	Pituitary
Hardest substance	Tooth enamel
Largest muscle	Gluteus maximus
Smallest muscle	Stapedius
Blood volume (adult)	5–6 litres
Heart beats per day	~1,00,000
Normal body temperature	37°C (98.6°F)
Brain weight	~1.4 kg
Neurons in brain	~86 billion
RBC lifespan	120 days
Platelet lifespan	7–10 days
WBC lifespan	Hours to years (type-dependent)
pH of blood	7.35–7.45

APPENDIX B — Scientific Names (Examiner Favourites)

Common name	Scientific name
Human	*Homo sapiens*
Tiger	*Panthera tigris*
Lion	*Panthera leo*
Leopard	*Panthera pardus*
Cheetah	*Acinonyx jubatus*
Indian elephant	*Elephas maximus*
African elephant	*Loxodonta africana*
Peacock	*Pavo cristatus*
King cobra	*Ophiophagus hannah*
Mango	*Mangifera indica*
Rice	*Oryza sativa*
Wheat	*Triticum aestivum*
Maize	*Zea mays*
Sugarcane	*Saccharum officinarum*
Tea	*Camellia sinensis*
Lotus	*Nelumbo nucifera*
Banyan	*Ficus benghalensis*
Neem	*Azadirachta indica*
Tulsi	*Ocimum tenuiflorum*
Onion	*Allium cepa*

APPENDIX C — Trap-Recognition Card

The Trap	The Truth
Universal donor = O+	NO → **O−** (the minus/Rh− is essential)
Largest cell = neuron	NO → Ostrich egg is largest; neuron is longest
Anopheles = dengue	NO → Anopheles = **malaria**; Aedes = dengue

The Trap	The Truth
Insulin from α-cells	NO → **β-cells** (Islets of Langerhans) produce insulin
Cardiac muscle is voluntary	NO → involuntary AND striated (unique)
Vitamin D deficiency = rickets only	Also **osteomalacia** in adults
Father of genetics = Darwin	NO → **Mendel**; Darwin = natural selection
Bile produced by gall bladder	NO → bile PRODUCED by liver; STORED in gall bladder
Largest endocrine gland = pituitary	NO → **Thyroid** is largest; pituitary is "master"
Smallpox eradicated by DDT	NO → **vaccination** (Edward Jenner's principle)

APPENDIX D — 25-Question Mini-Mock

Instructions: 25 questions, 25 minutes. No negative marking in this self-test.

1. Which organelle is called the "powerhouse of the cell"? (a) Nucleus (b) Ribosome (c) Mitochondria (d) Lysosome
2. Which organelle is called "suicide bags" due to its autolytic function? (a) Mitochondria (b) Lysosomes (c) Golgi body (d) Endoplasmic reticulum
3. Who is called the "Father of Genetics"? (a) Darwin (b) Watson (c) Mendel (d) Lamarck
4. The start codon in mRNA translation is: (a) UAA (b) UAG (c) AUG (d) UGA
5. Which blood group is the universal donor? (a) A+ (b) B− (c) O− (d) AB+
6. The largest bone in the human body is: (a) Tibia (b) Femur (c) Humerus (d) Radius
7. The smallest bone in the human body is: (a) Stapes (b) Malleus (c) Incus (d) Clavicle
8. The largest gland in the human body is: (a) Pancreas (b) Thyroid (c) Liver (d) Adrenal
9. Which gland is called the "master gland"? (a) Thyroid (b) Pineal (c) Adrenal (d) Pituitary
10. The "fight-or-flight" hormone is: (a) Insulin (b) Thyroxine (c) Adrenaline (d) Glucagon
11. Goitre is caused by a deficiency of: (a) Iron (b) Calcium (c) Iodine (d) Zinc
12. Which vitamin is essential for blood clotting? (a) Vitamin A (b) Vitamin C (c) Vitamin D (d) Vitamin K
13. Which vitamin is synthesised by the human body on exposure to sunlight? (a) Vitamin A (b) Vitamin B_{12} (c) Vitamin C (d) Vitamin D
14. Which vitamin is found exclusively in animal-source foods? (a) Vitamin C (b) Vitamin B_{12} (c) Vitamin D (d) Vitamin K
15. Beri-beri is caused by deficiency of: (a) Vitamin B_1 (Thiamine) (b) Vitamin B_2 (c) Vitamin B_6 (d) Vitamin B_{12}
16. The vector of malaria is: (a) Male Anopheles mosquito (b) Female Anopheles mosquito (c) Female Aedes aegypti (d) Culex mosquito
17. The vector of dengue fever is: (a) Male Anopheles mosquito (b) Culex mosquito (c) Female Aedes aegypti (d) Female Anopheles mosquito

18. The vector of filariasis is: (a) Aedes aegypti (b) Anopheles (c) Culex (d) Sandfly
19. The first disease to be eradicated globally through vaccination was: (a) Polio (b) Smallpox (c) Tuberculosis (d) Measles
20. India was officially declared polio-free in: (a) 2012 (b) 2013 (c) 2014 (d) 2015
21. The balance and coordination centre of the brain is: (a) Cerebrum (b) Medulla oblongata (c) Hypothalamus (d) Cerebellum
22. The largest part of the human brain is: (a) Cerebellum (b) Cerebrum (c) Medulla (d) Thalamus
23. The pyramid of energy in an ecosystem is: (a) Always inverted (b) Always upright (c) Upright only in aquatic ecosystems (d) Varies by season
24. The 10% energy rule in ecology means: (a) 10% of energy is lost to respiration; 90% passes on (b) Only 10% of energy at one trophic level transfers to the next (c) 10% of energy is stored as biomass permanently (d) 10% of producers are consumed by primary consumers
25. How many biodiversity hotspots does India have? (a) Two (b) Three (c) Four (d) Five

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Answer key: 1-c | 2-b | 3-c | 4-c | 5-c | 6-b | 7-a | 8-c | 9-d | 10-c | 11-c | 12-d | 13-d | 14-b | 15-a | 16-b | 17-c | 18-c | 19-b | 20-c | 21-d | 22-b | 23-b | 24-b | 25-c

Scoring: 23–25 = Excellent | 18–22 = Good | Below 18 = revise weak chapters before your mock series.

EXAM ALERT

Error-log rule: For every question you got wrong, write one sentence explaining what you missed. Retest yourself on only those questions 3 days later — without looking at the solution. This single habit doubles what the mock teaches you.

◎ CHAPTER SUMMARY — EVERYTHING IN ONE VIEW